Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 · UNIT 4 · CLASS 51.2 Moment of Inertia Numerical

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 51.2 Moment of Inertia Numerical

1 CLASS 51.2 Moment of Inertia Numerical

Study focus: CLASS 51.2 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000106 · EV0000107

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

So the first is moment of inner ship a rectangle about its centroid and x-axis.

Key details

- Let the height of this rectangle be h.
- Like the total area of this rectangle that is capital A will be equal to B into H.
- Next, we want to find inertia about the centroidal x-axis.
- First, we should mark the centroidal x-axis here.
- Where will be the centroidal x-axis?
- Then where will be the centroidal x axis?
- We know why by formula for rectangle is h by 2.
- The shaded area is the element.
- Since this is a small area, let it be dA.
- The distance is y.
- I x0 will be equal to we know moment of inertia's distance square into area.
- Here the distance is y, so y square into area is d.
- Next, this is the inertia of the smaller element about the centroidal x-axis.
- Next, I need to find the inertia of the rectangle about the centroidal x-axis.
- Once I integrate this, integral of d i x naught will be equal to i x naught.
- This is the moment of inertia of a rectangle above the centroid x axis.
- This will be equal to integral of y square into d.
- Integral of y square.
- into d a.
- What is d?
- D is the area of the small element.

- The area of the element will be equal to since it is in the form of rectangle, it will be equal to b into h.
- Here what is b?
- Since these two lines are parallel, the width of the element will also be equal to.
- B itself, so d a is equals to b into h.
- What is this h?
- The width of the element is a variable along y direction.
- Next, i x naught will be equal to integral of b's constant.
- Limits will be from C minus H by 22 plus H by 2.
- I x0 will be equal to B by 3.
- It will be equal to h by 2 4 cube minus of h by to cube.
- I x naught will be equal to b h cube by.
- This is the moment of inertia of a rectangular lamina above the centroid and x axis.
- That will be equal to bh cube by 12.
- You have to remember this.
- Moment of inertia of a rectangle about its centroidal x axis is b by 12.
- There is a rectangle like this, and here is the x-axis, and centroidal x axis is at a distance of.
- H by 2 from here.
- Moment of inertia about this axis, we got it as BH U by 12.

Example or application from the lecture

- For this I am going to consider a rectangle.
- Let us consider this rectangle of width B.
- If you consider this is just an x-axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

Core explanation · 05:00–09:59

We can apply parallel axis theorem.

Key details

- According to parallel axis theorem, inertia about any axis that is ix will be equals to inertia about parallel centroidal axis that is i bar x plus area into distance squared.
- This is the statement of parallel axis theorem.
- I bar X already we know B H cube by 12.
- Area is area of a rectangle that is B into H.
- Then D stands for the distance between the two parallel lines that is H by 2.
- Plus area is b into h, then the distance is h by 2 square.
- Just simplify this, we are going to get ix as this will be bh cube by 4, and this is bh cube by 12.
- [ASR low confidence; check audio].
- B.H.Q.
- B.H.Q.B.

- This is the moment of inertia of a rectangle above the centroidal, sorry, above the line passing through its base.
- Its bottom most member will be BHQ by three.
- Next is moment of inertia of a circle about its diameter axis.
- [ASR low confidence; check audio].
- Of radius R.
- Let O be the center of the circle.
- Let the radius of the circle let it be R.
- Because for circle, this is where the centroid.
- R0 with respect to center.
- Then the radial width of this element let it be DR0.
- Next, let the inclination of this element with respect to this centroidal x axis adjust the x-axis.
- Let it be theta, then the inscribed angle of this element let it be d theta.
- This is R0 and this is D R naught and this is D theta.
- Since it is like an arc and the arc this arc length will be equal to r naught into inscribed angle d theta.
- The area of this element it's on when D theta is small it means almost like a rectangle.
- This is the area of the smaller element.
- This is the inertia of the smaller element.
- Once we integrate this, we are going to get it as i x naught will be equal to integral of y square into d e.
- Next what is 1?
- D is R0 into DR0 into D theta.
- There are two variables.
- One, this element can be anywhere from 0 to R within this radius.
- At the same time, theta, this element can be anywhere from 0 to 2 pi.
- Y is nothing but this is the x-axis, it's like this, and this is y.
- When this d theta is very small, the element will be just like a point, then y will be the opposite side of this right angle triangle.
- This is theta, we know.

Example or application from the lecture

- For this I am going to consider a circle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 10:00–14:59

Zero two pine zero two or.

Key details

- Then R0 square into R0 will be equal to R0 Q into D R naught.
- This is the term related to R0, then sin square theta into D theta.

- Integration of R0 cube will be equal to R0 to the power of 4 by 4, and if you apply the limits from 0 to R, that term is going to be.
- R to the power of 4 by 4, one more term is anyhow 0.
- Then we need to integrate 0 to 2 pi sine square theta into d theta.
- Sin square theta will be equal to 1 minus cos 2 theta by 2.
- And for this the limits will be from 0 to 2 pi.
- Now this is IX0 is equal to R0 to the power r to the power of 4 by 4, this is first term.
- Then integration of 1 is.
- I'll take this 1 by 2 outside, so it's going to be 1 by 2.
- Then integration of 1 is theta because it is with respect to into d theta here and down d theta is there.
- Cos two theta will be equals to sine two theta by two, limits will be from zero to two by.
- Next, this will be equal to R to the power of four by eight.
- substitute it as two pi, two pi, then minus of sine of two times it of two pi divided by two minus of zero and zero.
- This will be equal to 5 R to the power of 4 by 8.
- This is the moment of inertia fit circle about the diameter axis.
- I x naught will be equal to i y naught.
- I can write it as i bar x is equals to i bar y is equals to pi r to the power of 4 by.
- See this is the moment of inertia of a circle about the diameter x-axis as well as the y axis.
- This is about circles.
- Only difference is the limits will be varying from zero to pi instead of two pi.
- here you can see the formulas for the semicircle.
- If you just derive it just like a circle by changing the limits from 0 to pi, you are going to get the values.
- Here, moment of inertia of a circle, sorry, semicircle, which is symmetrical about y-axis here.
- try to learn the formulas this way.
- Inertia about an axis of symmetry and the diameter axis is always equals, that is pi r to the power of 4 by 8 and pi r to the power of 4 by 8.
- Here, this is the centroid which is at a distance of 4 r by 3 pi.
- If you draw a horizontal line, this is the centroidal axis which is parallel to diametral axis.
- Moment of inertia about this line is 0.11 into R to the paraph.
- In that case, inertia about diameter and axis of symmetry is pi r to the power of 4 by 8, and the centroidal axis parallel to the diameter line is 0.11 into r to the power of 4.
- That's about semicircle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 15:00–19:59

In the derivation of circle, we are going to get ix and iy value.

Key details

- One is pi r to the power of 4 by 16, and this is also pi r to the power of 4 by 16.

- This is moment of inertia of a quarter circle about the radial lines.
- Next moment of inertia of a quarter circle.
- About the centroidal x axis and centroidal y axis is point five fi into r to the power of four.
- That's it for today's class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 51.2 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.2 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.2 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 51.2 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

C Revision and retrieval practice